INFO3315 Part I Report

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1 Introduction

1.1 Background and Context

Food waste is one of the central issues addressed in the United Nations' Sustainable Development Goals (SDGs). This report specifically focuses on SDG 12: Responsible Consumption and Production which highlights the importance of ensuring sustainable patterns of consumption and production across all sectors of society. SDG 12 calls for halving per capita global food waste at the retail and consumer levels, as well as reducing food losses along production and supply chains by 2030 (United Nations, 2025). This global commitment reflects the recognition that wasteful practices in the food cycle directly undermine environmental sustainability.

1.2 Aim of Investigation

Within this global framework, our project focuses on reducing food loss at the consumer stage. Consumers frequently face difficulties in tracking what food they own will expire, and how to use ingredients efficiently before they spoil. Without intervention, these small but frequent losses accumulate into significant volumes of waste which undermine the achievement of SDG 12.

Our proposed solution is motivated by the recognition that changes in the consumer stage are essential for environmental change. A mobile application tailored for consumer use can bridge this gap by providing tools to monitor food inventory, receive timely reminders about expiry dates, and generate recipes based on available ingredients. By designing such a system, we aim not only to reduce unnecessary waste but also to cultivate sustainable consumption habits among individuals and families.

1.3 Competitor Analysis and Gaps in Existing Solutions

Background research was conducted. Our group examined No Waste Australia which is a major influence in the food waste reduction industry. This company primarily specialises in handling surplus stock from businesses, storing, redistributing, and repurposing it for further use (No Waste Australia, 2025).

Existing Solution (No Waste)	
Advantages	**Disadvantages**
Sustainability-first approach.	Consumer-centre approach.
Established reputation in the food redistribution sector.	Consumer neglect: No major applications currently target the household level, even though consumers represent a large share of the waste problem.
Strong logistical and storage capabilities.	Transparency issues: Organisations like No Waste Australia are not highly transparent about

	internal processes, which can create trust barriers for customers.
Massive technical and financial support.	Technological barriers: Without integration of consumer tools such as barcode scanning, AI image recognition, or community sharing features, it is difficult to motivate individuals to change their behaviours.
A wide network of partnerships with retailers and charities.	Behavioral challenges: Consumers value autonomy and privacy. Any solution that feels intrusive or time-consuming risks rejection. Any suggestions that invade personal privacy may also risk rejection.
flexibility in managing a broad variety of products, which strengthens their ability to repurpose goods and support businesses in reducing unnecessary waste (No Waste Australia, 2025).	n/a

These limitations highlight the need for a consumer-centre technological solution. Thus our project fills this gap by designing a user-friendly mobile application that focuses on helping individuals manage food and connects them to broader sustainability efforts.

1.4 Stakeholders and Users

A PACT analysis was undertaken to frame the design considerations. From a stakeholder perspective, the solution involves:

- Government and NGOs – who are invested in addressing food insecurity and meeting the overall sustainability goals.
- Supermarkets and retailers – who benefit from reduced inventory waste and better procurement planning.
- End consumers (households and individuals) – who can gain financially and environmentally from reduced food loss.

The core users of the application are everyday shoppers and household managers. These users need support in planning purchases, monitoring food expiry, and making the most of their ingredients.

1.5 Research Questions

To guide the design of the application, the following research questions frame our investigation:

1. What challenges do consumers face in monitoring food inventory and preventing waste at home?

2. Which technological features (e.g., ripeness recognition, expiry reminders, community recipes) are most effective and acceptable to users?
3. How can the interface be designed to support quick, intuitive use in diverse contexts such as supermarkets and kitchens?
4. What kinds of incentives or social features can sustain user engagement and loyalty?
5. How can privacy and autonomy concerns be addressed while still leveraging data for useful insights?

These questions ensure that the project remains user-centred, aligning with both the requirements of SDG 12 and the specific expectations of this course.

2 Method

2.1 Survey

Our project favors the use of surveys to collect user requirements due to three major advantages: they enable the rapid acquisition of a large volume of feedback from geographically dispersed user groups, are cost-effective, and minimally intrusive. More importantly, we do not collect users' email addresses and adhere to strict anonymization of the collected data (see figure 2.1). This approach encourages users to provide truthful responses, thereby reducing the impact of subjective bias on the results.

2.1.1 Survey Conduction

- Define research objectives
  As our group has chosen "ensure sustainable consumption and production patterns" as the main problem to solve and plan to design a mobile APP to inform people to save food waste. So we plan to gather insights from potential users regarding their experiences storing fresh food in this survey, identify potential causes of food waste, and explore possible solutions to address the issue.

- Identifying the target audience
  Given that our potential users are urban residents, our solution primarily addresses the challenges of supermarket shopping and food storage. To further narrow down our demographic, we have identified university students as the primary group of respondents. This segment is highly relevant because students often face unique constraints such as smaller living spaces, tighter budgets, and irregular schedules, all of which amplify the difficulties of grocery shopping and food preservation. By targeting this group, we aim to gather insights that reflect the real world pain points of young urban dwellers, enabling us to design a solution that is both practical and impactful.

- Design the survey
  For survey question design, we start with simple, broad questions and gradually move toward specific, sensitive ones. The beginning questions are relevant to the responder's daily life (see figure 2.2). Beside that, we design both open-end and close-end questions, ensure that the types of questions are diverse and cover a

sufficient number of aspects. In designing, we avoid using leading or biased terminology to ensure questions are clear and unambiguous and keep the survey concise to prevent user fatigue and maintain data quality.
Based on the studies about the relationship between impulsivity, food storage and waste (O'Neill et al., 2024). We especially want to focus on impulsivity and storage (Etim et al., 2025). As a matter of fact, in this survey we gain a comprehensive understanding of respondents' shopping habits and storage habits, we design the 11 questions (survey link) to realize the respondent's shopping frequency, whether you routinely check expiration dates, whether it's difficult to track how long food has been stored, whether you plan purchases/make lists of quantities, whether you make impulse buys, how often you discard leftovers after meals, the primary cause of waste and expected APP functions.

- **Validation of Survey Method**
  The sample originates from an online survey distributed via social channels and ED forum, therefore it's non-probability sampling. Furthermore, the relatively small sample size limits the statistical power of the findings. Affected by these factors, the sample lacks sufficient representativeness to support inferential strict analysis or to make broad generalizations about the entire population under study. While all respondents from universities in Sydney, the way questions are designed in the survey focusing on general shopping habits, we believe this bias is limited and it can be validly seen as a window into behaviours and interests for the general consumer system. However, a larger, and more diverse sample would be required to apply these insights more accurately to be used for real-world applications.

2.1.2 Data Analysis

- Data cleaning
  First, the dataset will be examined for unanswered questions or incomplete responses. Depending on the extent and nature of the missing data, we will consider sample deletion if the proportion of missing values is small and appears randomly distributed, the affected cases may be removed to maintain data integrity. Secondly, to facilitate statistical analysis, non-numeric responses will be converted into numerical formats. For example, binary responses such as "Yes" and "No" will be coded as 1 and 0, respectively. Similarly, categorical variables will be transformed using appropriate coding schemes to ensure compatibility with analytical tools. Finally, open ended responses will be manually reviewed to identify and remove meaningless or irrelevant answers, such as random characters, repeated letters, or responses that do not address the question.

- Descriptive statistics
  The result is collected and shown by pie chart.

- Text analysis (for open ended)
  Users expected functions: Freshness Identification/Scan/Photo Judgment (2), Expense/Sustainability/Waste Amount Tracking (2), Inventory/Overstock Prevention

(1), Portion Planning (1), Donation/Distribution (1), Cooking Guidance (1), with the remainder categorized as Invalid(4).

2.2 Interviews

Our project also utilised interviews for qualitative data collection. We collected information on user experiences, preferences and concerns regarding food waste and potential solutions. Interviews provided the opportunity for us to gain more insight and context from participants that we otherwise wouldn't get from surveys alone. Below is the outline of our interview methodology.

2.2.1 Interview Conduction

The target audience for our interviews is grocery shoppers who may live alone, or with family members.

The questions we used are open ended to encourage the participants to give detailed responses. The questions mainly focus on food waste habits, motivations, and concerns regarding potential solutions. We also want to understand any privacy concerns and how they perceive the idea of an app that integrates with their grocery shopping app.

We chose these interview questions:

1. How often do you have to throw away food that has gone bad? Do you think it's a genuine problem?
2. Do you currently do anything to keep track of when your food will expire?
3. What is the biggest factor when considering cooking a recipe, e.g. quick.
4. Do you have any concerns about an app being linked to your grocery purchases to help prevent food waste?

We recruited participants through social media and personal connections and given that interviews take much longer than a survey we successfully conducted 7 interviews. The interviews were conducted online using zoom and ranged from 5 to 10 minutes in length. During the interview the participants' responses to the questions were recorded.

- **Validation of the Interview Method**
  Our group holds an offline face to face interview with the responders and records the conversation (interview conversation log, also see figure 2.3 and figure 2.4). We will do so to ensure the authenticity, holding various separate interviews with different groups of people trying to avoid the impact of the bias. We will use the interviews to gather a wide range of perspectives, with participants of varying ages and living situations. Having a diverse range of participants in the survey will help us understand varying points and find better solutions that cater to different people's lifestyles, and minimize potential biases. Thus we also see the interviews as a valid research method.

2.2.2 Data Analysis

During the interview the responses are recorded and organised. The answers are then carefully reviewed to find common patterns and insights. Once the data is collected, we will analyse it to answer these questions:

- What do people think is the biggest barrier to them reducing their food waste?
- How do participants feel about the idea of an app managing their food data?
- What specific features do they want in an app that would make them more likely to use it?
- What concerns do people have?

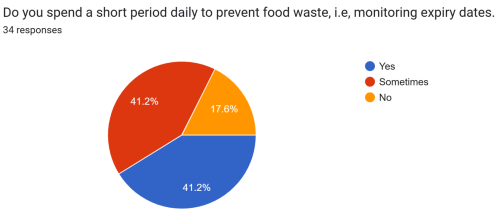
We will then use the findings from the interviews to combine with those from the surveys to build a more comprehensive view of our target audiences wants/needs and concerns. These findings will be used to inform the app's features and design.

3 Results & Discussion

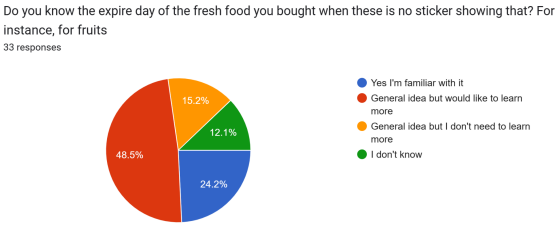
3.1 Survey

Shown below is a discussion of our results (aided by pie-charts) through our survey that will aid us as we develop our prototype.

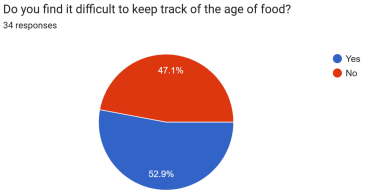
The crux of our issue in terms of the consumer level is that most people aren't monitoring expiry dates on a daily basis. As seen on the pie chart on the right, only 41.2% of our survey respondents always make a conscious effort to prevent food waste on a daily effort. Clearly there is an opportunity with our app to educate users the importance of minimizing waste and finding sustainable solutions; waste starts at the consumer level.



Another main issue that could be addressed through our application involves many having a clear lack of knowledge of fresh fruits expiry date. Our question here showed just 24.2% of survey respondents were aware of the expiry date of fresh fruits. The most common stance of respondents with 48.5% was that they had a general idea but there was still room to learn. More can be done on the app.



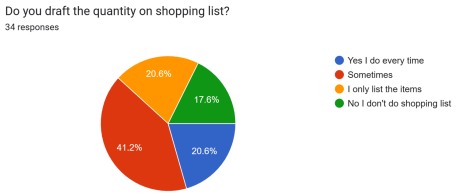
Alarmingly, despite clear guidelines on most foods of best before dates at supermarkets, the majority of consumers surveyed in the study still find it difficult to keep track of the age of their food. Again this highlights the necessity of



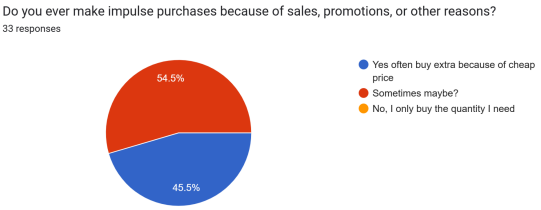
an inventory application and education through it to make it easier for users to keep track of the age of food.

Our application will need to consider a diverse plethora of explanations for wasted food ranging from problems of hectic lives, to buying ingredients but finding them too complex (possibly with overlap of busy lives) and changing meal plans. Clearly life is capricious, and for many, especially those in uni degrees hectic, and so our application will need to target ways to make it easier for shoppers to buy more sustainable or achievable amounts of food, and enable planning so they don't overestimate.

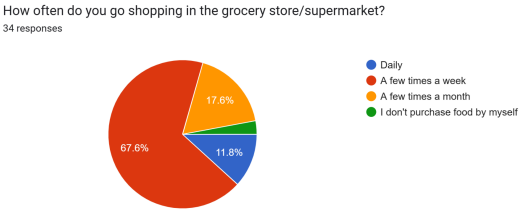
A significant finding was that only 20.6% of shoppers draft the quantity of their shopping list prior to every shop. This has huge ramifications; without a clear understanding of sufficient portions prior to shopping it is easier for shoppers to overestimate or make impulse purchases, which likely leads to waste for many cases.



And as seen on the pie chart here, every single respondent in the survey makes impulse purchases, either sometimes or always due to cheap prices. Our application could advise, provide a shopping list with proportions, and provide suitable recommendations so users buy sustainable amounts.



And with 79.4% of respondents visiting the grocery/store at least a few times a week, without a tool to advocate for waste-conscious purchasing people are just going to run into the same habits. An app to challenge the status quo is needed now.



We also asked on potential features:

For the Closed-Ended question, we devised four potential app features we thought useful for our app prototype to query with potential stakeholders. Inventory was the most useful at 35.3%. Also significantly high was the recipe generator at 32.4%. Though lower, a safe-to-use food checker did still account for 17.6% of survey responses, and educating users about safety of foods at 14.7%.

Due to the usefulness we aim to incorporate all four aspects, with a focus on the Inventory and Recipe Generator with higher proportions.

For the Open-Ended question, potential features we are interested in exploring further include some sort of food delivery, perhaps if it as excess so it can still be consumed, a portion plan, basic steps to cook, take a photo to test if food still safe-to-use, record waste to enlighten users to hold them accountable: perhaps rewards for good use we thought as well.

3.2 Interview

From the interviews we conducted, we gained many insights into peoples food waste habits, tracking methods and their views on different solutions. All of the participants stated that they think food wastage is a genuine issue for them. This was primarily caused by buying too much, forgetting food they had bought and not planning meals properly. Two out of the seven participants had strategies to reduce food waste such as freezing food.

For meal preparation most participants heavily prioritised convenience and how easy it was to clean up. This shows that a good solution is to generate quick easy recipes that use all of a person's food that is soon to expire. For privacy, participants were mostly open to integrating the app with grocery purchases as long as it had a strong privacy policy. This means when we design the app we must have transparent privacy settings and strong user control over their data.
Overall, the interviews showed many similar findings from the survey but allowed us to have a deeper understanding of people's motivations and concerns. Therefore our app should focus on being simple, convenient and privacy conscious.

4 User and System Requirements

For simplicity, as a result of our thorough analysis via surveys and interviews we summated the following User and System Requirements below to be used as a blueprint to guide us through further developments of our application throughout this project.

User Requirements

A summary of the various tasks we believe users will require out of our application are as follows. Here are our functional requirements as gathered from surveys alongside usability requirements gathered by interviews. When we have access to the software for the app we aim to expand into user experience requirements as well.

Functional Requirements from Surveys

- Using an inventory to view and monitor food, and age of food
- Keeping their inventory logged flexibly and correctly
- Education of fruit ages
- Generating recipes where ingredients are few and far between
- Putting ingredients through AI to check they are safe to use
- Posting new recipes

- Logging new supermarket shops
- Sustainable methods
    - Exploring different template shopping lists to not overbuy
    - Portion Plan: Shopping list makers with quantities
- Keep users motivated
- Inform users of their waste

Usability Requirements from Interviews

- simple
- convenience
- transparent privacy settings
- strong user control over their data

System Requirements

Explicitly in order to account for these user requirements, our system will need to:

- Provide an inventory log via a database system that seamlessly handles both the logging of food and monitoring of age, and displays the inventory clearly to users via the front-end of the application
- Ensuring maneuverability of contents in this inventory log whereby foods can easily be added and removed to correctly reflect users contents, which will involve user interaction at the front-end to be collected and used to manipulate the database at the back-end.
- Provide functionality through machine learning algorithms to be able to view fruits and judge their age
- Provide clear front-end support to be able to educate users on foods and interact with new recipes
- Provide front-end support for users to be able to create shopping lists and store them in the database if adding foods to their inventories
- Machine Learning algorithms should be used to generate recipes when ingredients are limited
- Front-End access should be available for certain default shopping lists that are designed to be sustainable amounts to prevent overbuying
- Provide rewards for sustainable practice on the front-end, and inform regardless
- The interface should be simple through a suitable use of necessary frameworks such as Nielsen's Heuristics
- The interface should be convenient, functioning quickly with a clear structure for achieving tasks
- There should be strong security in the back-end to prevent privacy breaches as there is lots of sensitive data at stake
- The user should have options for their data management in the application and have informed consent of the processes involved in the back-end through the front-end.

5 Implications and Limitations

Implications

The disconnect between what people intend and what they actually do reveals a fundamental challenge in reducing household food waste. While 82.4% of respondents claim to monitor expiry dates, over half report difficulty tracking food age. This paradox exposes a critical truth: good intentions alone cannot overcome the barriers that lead to food waste. Indeed, the solution requires more than simple reminder apps or tracking tools: it demands a comprehensive behavioral scaffold that bridges the gap between motivation and execution.

Consider the daily reality users face. They experience multiple layers of cognitive burden: remembering expiry dates, tracking inventory, resisting impulse purchases, and managing portion planning. The finding that 100% of respondents make impulse purchases, despite varying levels of planning, demonstrates that rational planning tools cannot overcome environmental and psychological triggers. Time pressure compounds these challenges, with “hectic lifestyle” emerging as a primary waste driver for 36.4% of respondents. The application must therefore function as a cognitive aid that reduces rather than adds to mental load.

The variety in waste prevention habits, with an equal number of respondents monitoring expiry dates daily versus sometimes, indicates that one-size-fits-all approaches will fail. The application must implement a graduated engagement model that initially requires minimal user effort while progressively building sustainable habits. Rather than assuming users will actively engage with tracking features, the design should incorporate passive data collection methods and interventions that maintain high visibility of food status without demanding constant attention. Gamification elements like streak counters, achievement badges, and comparative analytics can provide positive reinforcement for behavior change, acknowledging that transformation is gradual.

The distribution of feature preferences, with 35.3% prioritizing inventory management and 32.4% valuing recipe generation, suggests these functionalities shouldn’t exist in isolation. Deep integration between core features creates cascading interventions - for example, inventory tracking can adjust shopping lists, which then shape recipe suggestions and influence future purchasing, creating a virtuous cycle of waste reduction. This requires a unified data model where inventory status drives recipe suggestions in real-time, employing predictive analytics to anticipate spoilage risks and proactively suggest meals that prioritize at-risk ingredients.

The contradiction between claimed knowledge about fresh food expiry (72.7%) and reported difficulty tracking food age (52.9%) exposes a knowledge illusion where users overestimate their competence. Despite evident knowledge gaps, the limited interest in educational features (14.7%) reveals a critical insight: users resist explicit learning frameworks. As a result, education must be embedded within functional features rather than presented as standalone content. The system should implement experiential learning through real-time feedback loops, contextual micro-learning moments triggered by user actions, and comparative analytics that reveal misconceptions through data rather than instruction.

The interconnected nature of food waste drivers, where planning failures lead to impulse purchases, creating excess inventory and resulting in waste, demands a systems approach rather than isolated solutions. The application must function as an intelligent agent that identifies patterns users cannot see, suggests interventions they wouldn't consider, and creates conditions for success they couldn't establish independently. This positions the app not as a tool but as a behavioral partner that continuously evolves with user needs, learning from both successes and failures. Through this comprehensive approach, the app can finally bridge the persistent gap between intention and action in household food waste reduction.

Limitations

Our study faces significant sample limitations that restrict how broadly we can apply our findings. We collected only 34 responses through convenience sampling on social media (Instagram) and university forums (Ed Discussion), a sample too small for making claims about the wider population. The participants were mainly university students from urban Sydney, meaning the results may not reflect older adults, different income groups or rural communities with different food systems. People who choose to answer a food waste survey probably care more about the issue than average.

The methods we used also created constraints on our results. Some participants gave invalid answers to open-ended questions, which reduced the insights we could gather about feature development. Beyond these response quality issues, we faced deeper challenges with self-reporting. While participants expressed preferences for certain app features, wanting something doesn't mean using it. We couldn't verify what people actually do versus what they say they do. For instance, 88% reported wasting food less than once a week. Since we collected data at just one point in time, we also missed how shopping and waste patterns change with the seasons, when different foods become available and consumption habits shift.

Ultimately, our findings can be seen as a mirror into an incredibly thought-provoking issue and can certainly lead to meaningful interpretations on the state of food waste.

Appendix

Sign in to continue

To fill out this form, you must be signed in. Your identity will remain anonymous.

Report Abuse

SIGN IN

Figure 2.1

Do you spend a short period daily to prevent f	How often do you go shopping in the grocery

Figure 2.2

Interview

Key points:

Do you waste food, 18% of food wasted at the consumer level?

Uni-life is very hectic, makes it really tough to schedule,:
I think I have a larger number than 18% wastage! I feel motivated to cook something, but later on i fall a bit sick or tired from uni stuff, eventually i get other stuff, uber eats, around 25-26% wastage. Another factor is I often over-estimate the amount of food I can consume.

To combat wastage, people buy less more frequently:
I only buy a tiny bit of food for each meal, regularly in a week one bottle of drink. Less meals, more shopping, stock up on drinks. Milk goes off quickly.

Do you think standard recipes would be useful, or recipes with remaining ingredients, having an inventory to monitor stock?
Can have some naive stuff, hard-coded guidance notes. If you have actual ability to monitor crazy machine learning stuff, to actually evaluate, one can realistically consume, it would be hugely beneficial.

It can be quite annoying to record on the phone (which led me to say:)

Do you use notes? I feel there could be more suitable solutions to aid users.
I agree. A more interactive app than notes could definitely be useful.

The participant pointed out a huge issue with the potential platform: how to keep user motivated? It's a big problem.

Good way to log would certainly be a very interesting solution for people

Figure 2.3

Figure 2.4

Will - 6th Sept

1. How often do you have to throw away food that has gone bad? Do you think it's a genuine problem?

Often throws out food due to not cooking it before it expires or buys too much food. It's a genuine problem for him.

2. Do you currently do anything to keep track of when your food will expire?

He tries to estimate how much food he needs to buy but often overestimates

3. What is the biggest factor when considering cooking a recipe, e.g. quick.

Convenience of cleaning.

4. Do you have any concerns about an app being linked to your grocery purchases to help prevent food waste?

Depends on the privacy policy, for example if the data was sold he wouldn't want it linked.

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